

NumPy Proficiency Lab

From Array Creation to Machine Learning Patterns

Based on: *1.Numpy_Fundamental.pdf*

40 Hands-on Programming Questions (No Theory)

 Solve by coding only — no written explanations. Each question tests a specific NumPy concept.



BASIC: Array Creation & Properties

? **Question** Create a 1D NumPy array from the Python list [10, 20, 30, 40, 50] and print its type.

? **Question** Generate a 2D array of shape (3,4) filled entirely with zeros using a NumPy function.

? **Question** Create a 3x3 identity matrix using a single NumPy function.

? **Question** Generate an array of 10 evenly spaced numbers between 0 and 1 (inclusive) using `np.linspace`.

? **Question** Create a 2x3 array of random integers between 5 and 15 (inclusive).

? **Question** Create a 3D NumPy array of shape (2,2,2) filled with the number 7 using `np.full`.

? **Question** For an array `arr = np.array([[1, 2, 3], [4, 5, 6]])`, write code to print its `shape`, `ndim`, `size`, and `dtype`.



BASIC to INTERMEDIATE: Indexing, Slicing & Reshaping

? **Question** Given `arr = np.array([10, 20, 30, 40, 50, 60])`, use slicing to extract [30, 40, 50].

? Question From a 2D array `matrix = np.arange(1, 13).reshape(3, 4)`, extract the last two rows and the first three columns.

? Question Using boolean indexing, extract all elements greater than 5 from `arr = np.array([1, 5, 8, 2, 9, 3])`.

? Question Reshape the 1D array `np.arange(1, 13)` into a (4,3) matrix, then flatten it back to 1D.

? Question Use `reshape(-1, 1)` to convert `np.array([1, 2, 3, 4, 5])` into a column vector. Then use `reshape(1, -1)` to convert it back to a row vector.

? Question Create a 1D array of 12 elements. Reshape it into a 3D array of shape (2, 3, 2) without hardcoding the total number of elements.

BASIC to INTERMEDIATE: Vectorized Operations

? Question Create two 1D arrays `a = np.array([1, 2, 3])` and `b = np.array([4, 5, 6])`. Compute the element-wise sum, product, and division (`a / b`).

? Question Compute the square of every element in the array `arr = np.array([2, 4, 6, 8])` without using a loop.

? Question Given `arr = np.array([1, 4, 9, 16])`, compute the square root of each element.

? **Question** Compare the values of two arrays `a = np.array([1, 3, 5])` and `b = np.array([2, 3, 4])`. Create a boolean array indicating where `a` is greater than `b`.

? **Question** Element-wise multiply a 2x3 matrix by a 1x3 row vector. Explain the broadcasting that happens.

☰ INTERMEDIATE: Mathematical & Statistical Operations

? **Question** For `data = np.array([10, 20, 30, 40, 50])`, compute the mean, median, standard deviation, sum, min, and max.

? **Question** Normalize the 1D array `data = np.array([10, 20, 30, 40, 50])` to have zero mean and unit variance. (Subtract mean, divide by std).

? **Question** For a 2D array `matrix = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])`, compute the sum of each column (`axis=0`) and the sum of each row (`axis=1`).

? **Question** Perform matrix multiplication of `A = np.array([[1, 2], [3, 4]])` and `B = np.array([[5, 6], [7, 8]])` using the `@` operator.

? **Question** Compute the inverse of the matrix `A = np.array([[4, 7], [2, 6]])`.

? **Question** Solve the system of linear equations:

$$\begin{cases} 3x + y = 9 \\ x + 2y = 8 \end{cases}$$

using `np.linalg.solve()`.



INTERMEDIATE: Broadcasting

? Question Add the scalar 5 to every element of the array `arr = np.arange(1, 6)` using broadcasting.

? Question Create a 3x4 matrix and a 1x4 row vector. Add the vector to the matrix. Explain the broadcasting shape compatibility.

? Question Create a 3x4 matrix and a 3x1 column vector. Add the vector to the matrix.

? Question Normalize a dataset of shape (50, 10) (50 samples, 10 features) by subtracting the mean of each feature (shape (10,)) and dividing by the standard deviation of each feature using broadcasting.



INTERMEDIATE: Random Numbers & Data Splitting

? Question Set the random seed to 42 and generate a 2x3 array of random numbers from a standard normal distribution (mean=0, std=1).

? Question Create an array [1, 2, 3, 4, 5] and shuffle it in-place.

? Question Generate a random permutation of the numbers 0 through 9.

? Question Write code to perform an 80/20 train-test split on a dataset `X` of shape (100, 5) and labels `y` of shape (100,) using `np.random.permutation`.



ADVANCED: Array Manipulation

? Question Stack two 2x2 arrays vertically (`np.vstack`) and horizontally (`np.hstack`).

? Question Given a feature matrix `X` of shape (100, 3), add a bias column (column of ones) as the first column using `np.hstack`.

? Question Use `np.where` on `arr = np.array([1, 2, 3, 4, 5, 6, 7, 8])` to:

- (a) Get the indices where values are greater than 5.
- (b) Create a new array where values ≥ 5 are labeled "High" and others "Low".
- (c) Replace all values ≤ 5 with 0, keeping values ≥ 5 as they are.

? Question Combine `a = np.array([1, 2, 3])` and `b = np.array([4, 5, 6])` along a new axis (depth-wise) using `np.stack` to create a 2x3 array.



ADVANCED: Real-World / ML Patterns

? Question Implement the **Mean Squared Error (MSE)** formula using NumPy:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_{\text{pred},i} - y_{\text{actual},i})^2$$

Assume `predictions` and `actual` are 1D NumPy arrays of the same length.

? Question Using the normal equation, compute the optimal parameters `theta` for linear regression:

$$\theta = (X^T X)^{-1} X^T y$$

Assume `X` is a (100, 3) feature matrix and `y` is a (100,) target vector.

? Question Convert the label array `labels = np.array([0, 1, 2, 1, 0, 2])` into a one-hot encoded matrix of shape (6, 3) using `np.eye`.

? Question Write a function that processes a large dataset `data` of shape (1000, 50) in batches of size 64 using a for-loop and NumPy slicing. Inside the loop, just print the shape of each batch.

✔ Ready to Code! ✔

“Vectorization is your superpower — never write a Python loop for array operations again.”

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